

Loan Status Analysis-Insights

Loan Portfolio Overview

- **Total Applications:** 45,000
- **Approved Loans:** 10,000 (~22%)
- **Rejected Loans:** 35,000 (~78%)
- **Average Loan Amount:** 9,583
- **Average Interest Rate:** 11.01%

 **Interpretation:** The approval rate is relatively low, suggesting either strict credit policies or a high-risk applicant pool. The average loan amount and interest rate indicate moderate lending terms.

Education & Loan Approvals

- **Bachelor's Degree:** 30.18% of approvals (~3,000 loans)
- **High School:** 26.71% (~3,000 loans)
- **Associate Degree:** 26.5% (~3,000 loans)
- **Master's Degree:** 15.19% (~2,000 loans)
- **Doctorate:** negligible approvals

 **Interpretation:** Mid-level education groups (Bachelor, High School, Associate) dominate approvals. Higher education (Master/Doctorate) does not significantly improve approval chances, possibly due to income-to-debt ratios or risk perception.

Home Ownership & Loan Status

- **Renters & Mortgage Holders:** Largest share of applications, with high rejection rates.
- **Owners:** Smaller applicant pool but relatively better approval ratios.
- **Others:** Minimal representation.

 **Interpretation:** Housing stability (ownership) correlates positively with loan approval, while renters and mortgage holders face higher rejection risk.

Gender & Default History

- **Gender:** Male applicants slightly outnumber females in both approvals and rejections.
- **Default History:** Applicants with prior defaults face disproportionately high rejection rates regardless of gender.

 **Interpretation:** Default history is the strongest predictor of rejection. Gender differences are minor compared to credit history impact.

Strategic Insights

1. **Approval Rate Challenge:** At 22%, the institution may be missing growth opportunities. Reviewing criteria could balance risk and expansion.
2. **Education Factor:** Mid-level education borrowers are more successful, but higher education does not guarantee approval—suggesting income stability matters more than qualifications.
3. **Housing Stability:** Ownership is a positive signal for lenders. Renters and mortgage holders may need additional risk assessment models.
4. **Credit History Dominance:** Past defaults overshadow other factors, reinforcing the need for robust credit scoring.
5. **Demographic Balance:** Gender distribution is fairly even, indicating no major bias in approvals.

Borrower's Risk Insights

Loan Amount vs Income

- Scatter plot shows wide variation: some borrowers with **low income request high loan amounts**, while others with high income request relatively modest loans.
- Outliers exist where loan requests are disproportionately large compared to income.

 **Interpretation:** Borrowers with mismatched loan-to-income ratios represent higher risk. Lenders should flag cases where loan requests exceed sustainable repayment capacity.

Loan Amount vs Loan Percent of Income

- Most borrowers fall within **0.1–0.3 loan-to-income ratio**, but some reach **0.5–0.6**, indicating heavy debt burden relative to income.
- Higher ratios correlate with increased rejection likelihood.

 **Interpretation:** Loan percent of income is a critical risk metric—borrowers with high ratios are more likely to default and face rejection.

Experience vs Loan Status

- **Approved borrowers** average more years of work experience (~5–6 years).
- **Rejected borrowers** average fewer years (~2–3 years).

 **Interpretation:** Work experience positively influences approval chances, as it signals career stability and repayment ability.

Credit History vs Loan Status

- Borrowers with **longer credit histories (10+ years)** show higher approval rates.
- Shorter credit histories (<5 years) are strongly associated with rejections.
- Approval likelihood rises steadily with credit history length.

 **Interpretation:** Credit history length is a decisive factor—longer histories build trust and reduce perceived risk.

Key Takeaways

1. **Income-to-loan ratio** is a strong rejection driver; sustainable ratios improve approval chances.
2. **Work experience** boosts approval likelihood, highlighting the importance of employment stability.
3. **Credit history length** is one of the most influential factors—borrowers with longer records are favored.
4. **Risk hotspots:** borrowers with high loan-to-income ratios, short credit histories, and limited work experience face the highest rejection risk.